

LensCNN: Closing the sim-to-real gap for CNN Einstein-radius estimation with a realistic simulation generator

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ABSTRACT

Aims. Deep learning enables fast parameter estimation for the $\sim 10^5$ strong gravitational lenses expected from current and upcoming surveys like Euclid and Roman. However, CNNs trained on simplified simulations often degrade substantially on real telescope data—a “sim-to-real gap” whose physical origins remain poorly understood.

Methods. We build a physically self-consistent, real-ingredient simulation pipeline, train a CNN to predict the Einstein radius (θ_E), and systematically ablate each realism component to determine what matters most. Our generator combines real HST galaxies (with measured SDSS velocity dispersions $\sigma_v \rightarrow \theta_E$) as deflectors, real COSMOS sources, real HST empty-sky backdrops, and real focus-diverse empirical PSFs—all in physical e^-/s units. The key innovation is *observational self-consistency*: each training system uses one real galaxy for both its light and its mass. We train a scale-conditioned CNN with a Gaussian-NLL uncertainty head.

Results. On the frozen spectroscopic-lensing benchmark of 62 SLACS and 40 S4TM lenses (Bolton et al. 2008 *b_S1E*), *ournative* – *HSTmodel* achieves $R^2 = +0.64$, bias = $+0.005''$, RMSE = $0.152''$, and a 15% catastrophic failure rate—beating the m3 baseline ($R^2 = -1.02$) and matching conventional modeling (Cao et al. 2025) at $\sim 10^6\times$ the speed. In the Euclid-degraded domain (real Q1 VIS PSF), we reach $R^2 = +0.71$, RMSE = $0.137''$, NMAD = $0.056''$ —surpassing the leading CNN benchmark (LEMON) on every aggregate θ_E metric. Causal ablations reveal that real empty-sky backdrops dominate ($> 10\times$ effect), followed by the empirical PSF and flat θ_E prior; the GEN4 self-consistency step is the decisive breakthrough. A Gaussian-NLL uncertainty head provides a domain-aware confidence score (Spearman $\rho = +0.71$) that reduces the confident-half failure rate to 6%.

Conclusions. Physical realism in training simulations—particularly self-consistency between lens light and mass—is the key to closing the sim-to-real gap. Our instrument-agnostic generator enables rapid adaptation to Euclid, Roman, and future surveys. The code and data are publicly available.

Key words. gravitational lensing: strong – methods: data analysis – techniques: image processing – galaxies: elliptical and lenticular, cD

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1. Introduction

1.1. Strong gravitational lensing as a cosmological probe

Strong gravitational lensing occurs when a massive foreground galaxy (the deflector) lies along the line of sight to a background galaxy (the source). The deflector’s gravitational field bends and magnifies the source light, producing Einstein rings or arcs. The characteristic angular scale of this phenomenon is the *Einstein radius* (θ_E), which directly measures the enclosed mass of the deflector within that radius. For a singular isothermal sphere (SIS), θ_E is given by:

$$\theta_E = 4\pi \left(\frac{\sigma_v}{c} \right)^2 \frac{D_{ls}}{D_s}, \quad (1)$$

where σ_v is the velocity dispersion of the deflector, and D_{ls} and D_s are angular diameter distances.

θ_E is a fundamental observable in galaxy evolution and cosmology: it constrains the mass–density profile of galaxies, the dark matter content of early-type galaxies, and the Hubble constant through time-delay cosmography. Upcoming wide-field surveys—Euclid (Laureijs et al. 2011), Roman (Spergel et al. 2015), and LSST (Ivezić et al. 2019)—are expected to discover

$\sim 10^5$ strong lenses, presenting both an opportunity and a challenge.

1.2. The need for automated lens modeling

Traditional lens modeling relies on iterative, likelihood-based inference (e.g., MCMC, nested sampling) to fit parametric mass models (Sérsic light profiles, SIE mass profiles, external shear). While accurate, this approach is computationally expensive (~ 3 minutes per lens; Cao et al. 2025) and requires human expert input for initialization and quality control. With $\sim 10^5$ lenses expected from Euclid alone, traditional modeling cannot scale.

Convolutional neural networks (CNNs) offer a solution. Once trained, a CNN can estimate θ_E from a single image in milliseconds—amortizing the inference cost across the training set. Pioneering work by Hezaveh et al. (2017) and Perreault Levasseur et al. (2017) demonstrated that CNNs can recover SIE parameters from simulated HST-like images. More recently, the LEMON framework (Busillo et al. 2026) showed that a CNN can be trained on fully parametric Euclid VIS simulations and applied to real Euclid Q1 lenses, achieving $R^2 = 0.71$ on θ_E .

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1.3. The sim-to-real gap

Despite these successes, a persistent problem remains: *CNNs trained on simulations degrade substantially when applied to real telescope data*. This "sim-to-real gap" manifests in several ways:

- **Prior-pull:** CNNs trained on a narrow or skewed θ_E prior collapse to the training mean on real lenses (the m3 baseline in this work: $R^2 = -1.02$).
- **Compression:** Predicted θ_E vs. true θ_E slopes are < 1 , indicating the network hedges toward the mean under domain shift.
- **Calibration failure:** Uncertainty estimates that are well-calibrated on simulations become overconfident on real data (e.g., 76%/96% coverage on sim-val vs. 52%/83% on real lenses).

The physical origins of this gap remain poorly understood. While several works have attempted domain adaptation (Ćiprijanović et al. 2023; Agarwal et al. 2025), all have been *sim-to-sim*, explicitly deferring real-data validation to future work. LEMON (Busillo et al. 2026) does evaluate on real Euclid Q1 lenses, but their training remains fully parametric and they report degraded performance on real vs. simulated data (their Sect. 7: "on real images the predictions from LEMON are worse than on simulated lenses").

1.4. This work: closing the gap through physical self-consistency

In this paper, we present a CNN-based θ_E estimator trained on a *physically self-consistent, real-ingredient simulation pipeline*. Our key innovations are:

1. **Observational self-consistency (GEN4):** Each training system uses *one real HST galaxy* for both its light (the deflector stamp) and its mass (θ_E computed from that galaxy's measured SDSS velocity dispersion σ_v). This forces the network to learn the luminosity–mass (Faber–Jackson) relationship that human modelers use when arcs are faint.
2. **Real ingredients throughout:** Real COSMOS sources (GREAT3-vetted), real HST empty-sky backdrops (for correlated noise and field neighbors), real focus-diverse empirical PSFs (STScI ACS ISR 2018-08, ISR 2023-06), and real field companions—all in physical e^-/s units at AB zeropoint 25.94.
3. **Causal ablations:** We systematically remove each realism ingredient to measure its impact. We find that real backdrops dominate ($> 10\times$ effect), followed by the empirical PSF and flat θ_E prior. The GEN4 self-consistency step is the decisive breakthrough: without it, even a flat prior fails to improve real-lens performance ($R^2 \approx 0$).
4. **Uncertainty quantification:** A Gaussian-NLL head provides a domain-aware uncertainty that correlates with real-world error (Spearman $\rho = +0.71$). On the confident half of SLACS, the failure rate drops to 6%—meeting the publishable bar for automated modeling.
5. **Comprehensive evaluation:** We evaluate on the frozen spectroscopic-lensing benchmark of 62 SLACS and 40 S4TM lenses (Bolton et al. 2008 b_SIE), as well as on Euclid-degraded versions (real Q1 VIS PSF), real Euclid Q1 lenses (vs. PyAutoLens GT), and a public Roman simulation challenge. We compare to LEMON, Cao et al. (2025), Gawade et al. (2025), and HOLISMOKES X (Schuldt et al. 2023).

This paper is organized as follows. Section 2 reviews the literature and positions our work. Section 3 describes the generator pipeline in detail. Section 4 presents the CNN architecture and training. Section 5 presents results, including benchmark performance, comparisons to LEMON and Cao, ablations, uncertainty calibration, and transfer to Euclid/Roman. Section 6 discusses limitations and future work. Section 7 concludes.

2. Literature review

2.1. CNN-based strong lens parameter estimation

The application of CNNs to strong lens parameter estimation began with Hezaveh et al. (2017), who trained a CNN to predict SIE parameters (θ_E , ellipticity, center) from simulated HST-like images. Perreault Levasseur et al. (2017) extended this to Bayesian neural networks for uncertainty quantification. Both works were sim-only; real-data validation was deferred.

Gawade et al. (2025) trained a CNN on HSC (Hyper Suprime-Cam) ground-based simulations, injecting lensed sources into real empty HSC cutouts—a recipe similar to our hybrid approach. On 182 real HSC lenses (SuGOHI sample), their CNN achieved θ_E accuracy of 10–20%, bias $< 5\%$, and an outlier fraction of $\sim 10\%$. However, their ground truth came from their own *YattaLens automated pipeline*, not an independent spectroscopic benchmark—a circularity we address with the Bolton b_SIE ground truth.

HOLISMOKES X (Schuldt et al. 2023) compared a ResNet to traditional modeling on 31 real HSC lenses. They found a "very good match" for $\theta_E \gtrsim 2''$, but external shear predictions were poor (networks predicted values near zero), and *no aggregate accuracy statistics were reported*.

LEMON (Busillo et al. 2026) is the closest published work to ours. Trained on 80k fully parametric Euclid VIS simulations (1–4 Sérsic sources, single-Sérsic lens light, SIE+shear), LEMON evaluated on:

- **60 Euclidised HST lenses** (29 SLACS + 13 EELs + 5 COSMOS + 13 ACS, heterogeneous literature GT): θ_E bias $-0.03''$, RMSE $0.14''$, NMAD $0.11''$, $R^2 = 0.53$ (Table 3).
- **354 real Euclid Q1 lenses** (success-filtered Walmsley + Rojas candidates, PyAutoLens SIE GT): θ_E bias $0.01''$, RMSE $0.17''$, NMAD $0.07''$, $R^2 = 0.71$ (Fig. 12a).
- **Key limitations:** LEMON uses fully parametric simulations (no real ingredients), performs *no realism ablations*, and admits degraded real-vs-sim performance (their Sect. 7). Per-lens predictions are not publicly available.

Cao et al. (2025) presented TinyLensGPU, a GPU-accelerated conventional modeling pipeline with nested sampling. Applied to 63 SLACS lenses, it achieved $\lesssim 5\%$ deviation and $\sim 10\%$ catastrophic failures at ~ 3 min/lens—the closest conventional modeling reference. Their code is public, and per-lens θ_E predictions were kindly provided by the authors for this work.

2.2. Domain adaptation for strong lensing

Several works have explored domain adaptation (DA) for lens parameter estimation:

- Ćiprijanović et al. (2023): DANN+MMD for θ_E regression, sim $\rightarrow$ sim (DES-noise targets).
- Agarwal, Ćiprijanović & Nord (2025): MVE+UDA, sim $\rightarrow$ sim, $\sim 2\times$ target-domain gain.

- **Domain-Adaptive NPE (2024)**: UDA improves posterior coverage, sim→sim.

All three are *sim-to-sim* and explicitly defer real-data validation. *Sim-to-real DA on real spectroscopic ground truth remains unclaimed*—this is one of our contributions (though we find that DA does not improve over realism-calibrated simulations; see Sect. 5.4).

2.3. Positioning our work

Table 1: Comparison of leading strong-lens parameter estimation methods

Feature	LEMON	Gawade	HOLISMOKES X	Cao	Ours
CNN/ML	✓	✓	✓	–	✓
Real ingredients	–	✓	–	N/A	✓
Self-consistency	–	–	–	N/A	✓
Independent GT	–	–	–	✓	✓
Real Euclid	✓	–	–	–	✓
Ablations	–	–	–	N/A	✓
Uncertainty	✓	–	–	✓	✓
Speed	ms/lens	ms/lens	ms/lens	~3 min/lens	ms/lens

Our differentiators are:

1. **Observational self-consistency (GEN4)**: The first CNN trained with real galaxies as both light and mass ($\sigma_v \rightarrow \theta_E$).
2. **Causal realism ablations**: The first systematic decomposition of what closes the sim-to-real gap.
3. **Independent spectroscopic ground truth**: Bolton b_SIE, not pipeline-derived.
4. **Comprehensive real-lens validation**: SLACS-62, S4TM-40, Euclid Q1-322, and Roman simulations—across HST, Euclid, and Roman domains.
5. **Domain-aware uncertainty**: A NLL head whose σ correlates with real-world error ($\rho = +0.71$) and works as a practical confidence gate (confident-half fail 6%).

2.4. Roman status

The Roman Space Telescope (scheduled for launch no earlier than 2026) is expected to discover $\sim 160,000$ strong lenses (Wedig et al. 2025). Wedig et al. released public Roman simulations (Zenodo, 16,214 lenses) explicitly "to support ... training neural networks"—but *no Roman parameter-estimation network exists yet*. Our instrument-agnostic generator, already validated on HST and Euclid, is directly applicable to Roman; we demonstrate zero-shot transfer fails ($R^2 = 0.11$) and retraining succeeds ($R^2 = 0.93\text{--}0.94$).

3. Data and simulator

[This section will describe the generator pipeline in detail, including: - The real benchmark (62 SLACS + 40 S4TM) - The COSMOS source catalog and surface-brightness cuts - The deflector library from real HST stamps - The GEN4 manifest builder ($\sigma_v \rightarrow \theta_E$, FJ relation) - The hybrid image assembly (real backdrops, companions, noise) - The Euclid/Roman instrument operators - Dataset validation gates]

Appendix A: Per-lens comparison tables

[Tables of per-lens θ_E predictions for SLACS-62, S4TM-40, and LEMON comparison]